

Question number	Answer	Additional guidance	Mark
3(a)	• 8.5×10^5 (1)		(1)

Question number	Answer	Additional guidance	Mark
3(b)	An explanation that includes the following points: • (no increase in number of bacteria) because the cells had to {adjust to new environment / warm up / synthesise new enzymes} (1) • (increase in number of bacteria) as {conditions are suitable / nutrients available / pH appropriate / oxygen levels appropriate / appropriate temperature} (1)		(2)

Question number	Answer	Additional guidance	Mark
3(c)(i)	A description that includes the following points: • use serial dilutions (1) • sample {added to agar and spread / spread over agar / used to create a lawn of bacteria (on agar)} (1)	ACCEPT description of serial dilutions e.g. dilute several times, / add bacteria to water and repeat	(2)

Question number	Answer	Additional guidance	Mark
3(c)(ii)	An explanation that includes three of the following points: • {optical methods / turbidity} measure {total cell counts / dead and alive cells} (1) • dilution plating measure (numbers of) living bacteria (1) • therefore in the optical method number of cells stayed the same as no new cells being produced (1) • therefore in the dilution plating number of cells stayed the same as number of new cells equals cells dying OR therefore in the dilution plating number of cells fell as were dying (1)		(3)